

Reference + Reframing

PARTIAL SUM VECTORS

for $1 \leq t \leq n$, an observation O_t is vector w/ length t

$$O_t = (s_1, s_2, \dots, s_t) = (\pi_1 + \sigma_1, \pi_2 + \sigma_2, \dots, \pi_t + \sigma_t)$$

REFRAMING

$$A_t(n) = |\{O_t \mid \pi, \sigma \in S_n\}|$$

IDENTITIES

$$\sum_{i=1}^n s_i = n(n+1)$$

HMM FRAMEWORK

Let $U_{\pi}^{(t)} \subset [n]$ and $U_{\sigma}^{(t)} \subset [n]$ track which numbers have been used up to step t

$$h_t = (U_{\pi}^{(t)}, U_{\sigma}^{(t)}) \Rightarrow |U_{\pi}^{(t)}| = |U_{\sigma}^{(t)}| = t$$

$$H_t = \{\text{all possible } h_t \mid O_t\}$$

$$T(h_t) = \left\{ h_{t+1} : \begin{array}{l} U_{\pi}^{(t+1)} = U_{\pi}^{(t)} \setminus \pi_{t+1} \\ U_{\sigma}^{(t+1)} = U_{\sigma}^{(t)} \setminus \sigma_{t+1} \end{array} \right\}$$

$$|T(h_t)| = (n-t)^2$$